

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Calculate each of the following derivatives.

a) $g(t) = \cot^5(\ln(5t^2))$

b) $h(s) = \cos^4(2s)e^{s^2-3s}$

c) Compute the second derivative $P''(t)$ for the function $P(t) = e^{\arctan(t)}$.

d) $f(x) = \left(\frac{\ln(x^7)}{\arctan(2x)} \right)^9$

2. Show how you can find the derivatives of the following functions *without* the use of the Chain Rule.

a) $g(x) = \frac{x}{2x+1}$

b) $H(x) = \left(\frac{-x^4}{\sqrt[7]{2x}} \right)^7 + \tan \left(2 \sin^{-1} \left(\frac{3}{5} \right) \right).$

c) $f(x) = \frac{(x+1)^2 e^{x+1}}{3x-2}$

3. Attempt to re-derive the derivative for $y = b^x$.

4. Compute each of the following derivatives.

a) $f(x) = 3^x$

b) $f(x) = 2^{x \cos(x)}$

c) $f(x) = \ln(2)^{\frac{x}{e^{2x}+1}}$

5. Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ mx + b & \text{if } x > 2 \end{cases}$. Find values of m and b that make f differentiable everywhere.

3. In algebra we learn about the concepts of *even* and *odd* functions. A function is called *odd* if $f(-x) = -f(x)$ for all values of x . A function is called *even* if $f(-x) = f(x)$ for all values of x .
- a) Sketch the graph of a possible odd function. Sketch the graph of a possible even function. *Hint:* It may be helpful to start by plotting a few points on each function and then thinking about what the definitions tell you about other points that must also be on these functions.
- b) Show that $y = x^2$ is even. Let n represent any *integer* value. Is $y = x^{2n}$ even for all values of n , odd for all values of n , or does it depend on n ? Justify your answer. What about $y = x^{2n+1}$?

- c) Show that the derivative of an *arbitrary* odd function is even and that the derivative of an *arbitrary* even function is odd.
- d) Show that if $f(x)$ is odd, then $f(0) = 0$. Is the same true for even functions?
- e) Give an example of a function that is both odd and even. Prove that your example is the only function that is both odd and even (*Hint: Consider what would have to be true if another function $g(x)$ which does not equal 0 for all values of x was both even and odd.*).